

Sajeesh K

+91-8590436272 sajeeshk234@gmail.com linkedin.com/in/sajeeshk1 github.com/sajee-sajee

Education

Bachelor of Technology — Computer Science & Engineering / *Lovely Professional University* Punjab, India Aug 2022 – Present

- Relevant coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, Computer Vision

Intermediate (Class XII) / *A.K.N.M.M.A.M HSS* Palakkad, Kerala Aug 2019 – Mar 2021

- Percentage: **90%**

Technical Skills

Languages	Python, C++, Java, JavaScript
ML / AI	TensorFlow, PyTorch, scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, HSV segmentation, deep learning (CNNs), data augmentation
Backend	Node.js, Express.js, Flask, Socket.IO, WebSocket, RESTful APIs
Frontend	React, Tailwind CSS, HTML5, CSS3
Databases	PostgreSQL, MySQL, MongoDB, SQLite
DevOps / Cloud	Docker, Azure (App Service, Functions, Blob Storage, Key Vault, Monitor, DevOps), AWS (core services, serverless, IAM), ARM Templates, CI/CD, Linux, Git, GitHub
AI / GenAI	LLMs, prompt engineering, generative models, diffusion models (Coursera certified)
Tools	Monaco Editor, Jupyter Notebook, VS Code

Experience

Student Ambassador / *LetsUpgrade EdTech Platform* Bangalore, India Jul 2024 – Aug 2024

- Organized and facilitated technical workshops on web development, cloud computing, and machine learning for **200+ students**, driving measurable engagement with emerging technologies.
- Led hands-on coding sessions covering React.js, Node.js, and Python; achieved an **85% workshop completion rate** through structured pacing and real-time Q&A support.
- Mentored students on full-stack development projects and conducted code reviews, accelerating participants' transition from concept to working prototypes.
- Authored technical tutorials and content on emerging technologies, collectively reaching **5,000+ students** across online platforms.

Azure Cloud Computing Intern / *Verzeo* Remote Jan 2023 – Mar 2023

- Completed an intensive training program on Microsoft Azure services — App Service, Azure Functions, Blob Storage, and Virtual Machines — with focus on security, scalability, and cost optimization.
- Deployed and managed production-grade web applications on Azure, implementing **Azure Key Vault** for secure credential management and **Azure Monitor** for real-time performance tracking.
- Built and maintained CI/CD pipelines using **Azure DevOps**; provisioned cloud infrastructure via ARM templates following infrastructure-as-code (IaC) best practices.

Projects

Automated Yield Estimation via Computer Vision / *Python · OpenCV · NumPy · Pandas · Matplotlib* Feb 2025 – Apr 2025

- Architected an end-to-end computer vision pipeline for automated fruit counting in orchards, eliminating manual sampling and reducing estimation time by an estimated **60%**.
- Implemented HSV color segmentation and contour detection with OpenCV to isolate and count fruit instances under variable lighting, achieving **>90% detection accuracy** on validation sets.

- Applied data augmentation (flipping, rotation, brightness jitter) to improve model robustness across real-world field conditions; deployed a lightweight inference variant optimized for low-resource edge environments.
- Engineered automated stock-logging and planogram compliance checks, cutting inventory reconciliation overhead by an estimated **40%**.

Health Monitoring System using Deep Learning / *Python · TensorFlow · Flask · SQLite ·* Sep 2024 – Nov 2024
HTML/CSS

- Built a real-time physiological monitoring system tracking heart rate, SpO₂, and body temperature from multi-sensor streams with **sub-second latency**.
- Designed and trained deep learning models in TensorFlow for sensor-fusion classification; implemented preprocessing pipelines that reduced signal noise by **~35%**, improving threshold-alert precision.
- Developed a Flask REST API backend persisting timestamped readings to SQLite and a live data-visualization dashboard enabling continuous multi-patient monitoring.
- Validated system performance using accuracy metrics, confusion matrices, and threshold-based alert evaluation across **5+ physiological edge cases**.

CollabRoom — Real-Time Collaborative Coding Platform / *React · Node.js · Socket.IO ·* Feb 2024 – Apr 2024
PostgreSQL · Docker

- Engineered a low-latency multi-user collaborative coding platform supporting **concurrent editing by 10+ users** with <100 ms synchronization lag via Socket.IO and WebSocket.
- Designed an operational-transform-style conflict-resolution layer on Node.js/Express.js to guarantee eventual consistency across distributed client sessions.
- Implemented sandboxed remote code execution for Python and JavaScript using **Docker containers**, isolating runtimes and enforcing resource limits to prevent host-system exposure.
- Delivered a responsive, dark-themed IDE experience using React, Tailwind CSS, and Monaco Editor; persisted user sessions and collaboration metadata in **PostgreSQL**.

Certifications

Machine Learning Specialization — Coursera (deeplearning.ai / Stanford Online)	Apr 2025
<i>/ Supervised & unsupervised learning, neural networks, recommender systems</i>	
AWS Developer Learning Plan — Amazon Web Services / <i>AWS core services, serverless, IAM, deployment best practices</i>	Dec 2024
Human-Computer Interface Design — NPTEL / <i>UX principles, usability testing, interaction design patterns</i>	Apr 2024
Introduction to Generative AI — Coursera (Google Cloud) / <i>LLMs, prompt engineering, diffusion models, generative applications</i>	Apr 2024
Data Structures & Algorithms — Lovely Professional University	Mar 2024
Full-Stack Web Development Bootcamp — Udemy / <i>HTML/CSS, JavaScript, React, Node.js, REST APIs, MongoDB</i>	Jun 2023
SQL for Data Science — Great Learning / <i>Joins, subqueries, window functions, data analysis workflows</i>	May 2023

Extracurriculars & Achievements

- Competed in **WEB Hackathon** (Mar 2024) — built and deployed a full-stack web solution under time constraints in a team setting.
- Attended **API Engineering Workshop** (Apr 2023) — hands-on REST/GraphQL API design and security best-practices session.
- Participated in **Open-Source Testing Tools Workshop** (Apr 2023) — automated testing, CI pipelines, and quality assurance fundamentals.
- Completed **Leadership Communication for Maximum Impact** (Mar 2023) — storytelling, cross-functional communication, and stakeholder presentation skills.